

moving video playback is a function of the response time and refresh rate of LCD and LED televisions. The type of TV backlighting has no effect on reproduction of fast-moving video.

Expected life. LED-backlit television is a new innovation that's not yet time-tested. However, LED TVs should score over CCFL-backlit LCD TVs in that LED lights last long and, unlike LCD TVs, don't change colour hue as much over time.

Sizes. LCD TVs are available in sizes ranging from 38 cm (15 inches) to 165 cm (65 inches). LED TVs are available between 116.8 cm (46 inches) and 177.8 cm (70 inches). Fabrication constraints of small-size LEDs restrict the size reduction of LED TVs. The use of Edge-LED lighting, however, makes the TV thinner than standard LCD TVs.

Power. LED TVs consume less power than LCD TVs of similar size.

Environmental effects. LED TVs are more environment-friendly, as mercury is not used in their manufacture.

Cost factor. LED TVs are expensive.

Q2. What is an OPC drum? How does a magnetic toner differ from a non-magnetic toner?

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A2. OPC stands for 'organic photoconductor.' The coating of the photoreceptor is made from carbon-based chemical compounds—specifically, photoconductive polymers synthesised from raw materials that are obtained by refining fossil fuels such as petroleum. OPC drums are environment-friendly photoreceptors available today because they utilise non-hazardous raw materials—and therefore a good alternative to hazardous photoreceptors such as arsenic triselenide (As_2Se_3) and selenium tellurium ($SeTe$) drums.

Physical characteristics. The most commonly used OPC drums receive a negative charge. From innermost to outermost layer, these typically

consist of an aluminium substrate, undercoat (or blocking) layer (UCL), charge-generation layer (CGL) and charge-transport layer (CTL). This description applies to most OPC drums in use today. Other types do exist, which utilise positive-charge OPC drums with a combined CGL and CTL (referred to as monolayer OPC drums). Positive-charge OPC drums typically have a shorter life than standard OPC drums, because their monolayer is less abrasion-resistant when incorporating softer materials usually confined to the CGL.

The aluminium substrate facilitates photoconductivity physically and electrically, but does not play an active role in the electrophotographic process. Its primary role is to provide structural and mechanical support, as well as an electrical path to ground.

The UCL acts as an interface between the substrate and photoconductive layers, to provide adhesion and prevent undesirable charge leakage that can adversely affect copy quality. Like the substrate, it does not play an active role in the electrophotographic process, but provides an electrical path to ground. Common UCL materials include aluminium oxide, anodised aluminium and various resistive polymers.

The CGL is extremely thin. Its colour, which typically determines the apparent colour of the OPC drum itself, depends on the specific materials that it contains. The light sensitivity of the CGL is a critical factor in OPC performance, and can be a limiting factor for the copy speed at which an OPC can function effectively.

The CTL is the outermost layer of an OPC drum. It is essentially transparent, allowing light to pass directly through to the CGL. Just as the CGL primarily determines the light-sensitivity of an OPC, the CTL primarily determines its charge-acceptance and charge-transport rate. As the outermost layer, the CTL is contacted by toner, developer, paper, drum-cleaning blade (or brush), ozone, and

other potentially abrasive and/or contaminating agents. Consequently, CTL wear characteristics, such as durability and abrasion-resistance, are critical factors in the potential life of an OPC drum.

Benefits of OPC technology. OPC drums are suitable for a wide range of applications, including copiers operating at extremely high speeds (75 cpm or above). These provide high levels of copy quality and long life formerly attainable only with As_2Se_3 drums. Also, OPC drums are not classified as hazardous waste. Manufacturing process is less expensive than the time-consuming, less efficient process required to manufacture selenium-based photoreceptors.

Toners are a chemical mix of ground plastic, lubricants and enhancers, used in laser printers/copiers for printing.

A non-magnetic toner having a single main ingredient without a carrier has two toner components of different mean grain sizes. The small-size toner component is less resistant than the large-size toner component, achieving a stable amount of electrified charge and thus an excellent image quality. A non-magnetic mono-component toner composed of a colouring powder contains a binder resin, colouring agent, wax and external additive.

Magnetic toners are formulated with a special grade of magnetic iron oxide that provides the characters with a magnetic signal. These are used in magnetic ink character recognition (MICR) technology. In the documents printed with standard (non-magnetic) toner, the characters appear to be correct but these are not magnetically readable by a reader-sorter. MICR technology is used in the banking industry as it allows fast and reliable document processing.

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